

Department of Computer Science

Applied Physics (Revised)

Credit Hour 3

Course Content:
Electric force and its applications and related problems, conservation of charge, charge quantization, Electric fields due to point charge and lines of force. Ring of charge, Disk of charge, A point charge in an electric field, Dipole in a n electric field, The flux of vector field, The flux of electric field, Gauss' Law, Application of Gauss' Law, Spherically symmetric charge distribution, A charge isolated conductor, Electric potential energy, Electric potentials, Calculating the potential from the field and related problem Potential due to point and continuous charge distribution, Potential due to dipole, equipotential surfaces, Calculating the field from the potential, Electric current, Current density, Resistance, Resistivity and conductivity, Ohm's law and its applications, The Hall effect, The magnetic force on a current, The Biot- Savart law, Line of B, Two parallel conductors, Amperes' s Law, Solenoid, Toroids, Faraday's experiments, Faraday's Law of Induction, Lenz's law, Motional emf, Induced electric field, Induced electric fields, The basic equation of electromagnetism, Induced Magnetic field, The displacement current, Reflection and Refraction of light waves, Total internal reflection, Two source interference, Double Slit interference, related problems, Interference from thin films, Diffraction and the wave theory, related problems, Single-Slit Diffraction, related problems, Polarization of electromagnetic waves, Polarizing sheets, related problems.
Teaching Methodology:
Lecturing, Written Assignments, Project, Experiments, Report Writing
Course Assessment:
Sessional Exam, Home Assignments, Quizzes, Experiments, Final Exam
Reference Materials:
1. Fundamentals of Physics (Extended), 10th edition, Resnick and Walker 2. Narciso Garcia, Arthur Damask, Steven Schwarz., "Physics for Computer Science Students", Springer Verlag, 1998

INTRODUCTION TO POLITICAL SCIENCE-I

03 Cr. Hrs

Objectives:

The objective of this course is to introduce the students with the fundamentals of the subject of Political Science and prepare them for advanced studies in the forthcoming semesters. The very basic concepts and terminology commonly used in the further courses of studies are taught to make the students friendly with the subject.

Contents:

1. Definition, Nature, Scope and Sub-fields of Political Science.
2. Relationship of Political Science with other social sciences.
3. Approaches to the study of Political Science: Traditional and behavioral approach.
4. State: its origin and evolution; Western and Islamic concepts of State,
5. Nation and Sovereignty.
6. Basic concepts of Political Science: Power, Authority, Legitimacy
7. Organs of Government: Legislature, Executive, Judiciary.

Q1 Define and explain electric charge, hence differentiate between conductors and insulators?

ELECTRIC CHARGE

The strength of electrical interaction of a particle with objects around it depends on its electric charge. Charges with same sign repel each other and charges with opposite signs attract each other. The object with equal amounts of the two kinds of charge is called electrically neutral. The object with imbalance amount of the two kinds of charge is called electrically charged.

All materials including solids, liquids and gases contain two basic particles of electric charge called electron and proton. The electron has negative polarity whereas proton has positive polarity.

It is the arrangement of electrons and protons that determines the electrical properties of all the substances. The sharing of electrons takes place when one material is rubbed against the other. The material that loses the electrons is called positively charged. The material that gains the electrons is called negatively charged.

EXAMPLE

"The rod gets positively charged when it is rubbed against the silk whereas silk is charged negatively.

UNIT OF CHARGE

"The SI unit of charge is called coulomb. The charge of 6.25×10^{18} electrons or protons is equal to one coulomb charge. The submultiples units of charge are μC , nC and pC etc.

Coulomb's Law

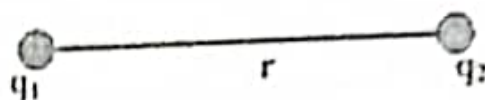
The electric force between two point charges is directly proportional to the product of the magnitudes of the charges and inversely proportional to the square of the distance between them.

EXPLANATION

Consider two point charges q_1 and q_2 separated by distance r . The magnitude of electric force F_e according to Coulomb's law is

$$F_e \propto q_1 q_2$$

$$F_e \propto \frac{1}{r^2}$$



On combining together

$$F_e \propto \frac{q_1 q_2}{r^2}$$

$$F_e = k \frac{q_1 q_2}{r^2}$$

Where k is constant of proportionality and depends upon the nature of medium between charges and system of units used to measure force, charge and distance. If medium between charges is free space, value of k is written as

$$k = \frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ N-m}^2/\text{C}^2$$

Where ϵ_0 is an electrical constant called permittivity of free space having value $\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2/\text{N-m}^2$.

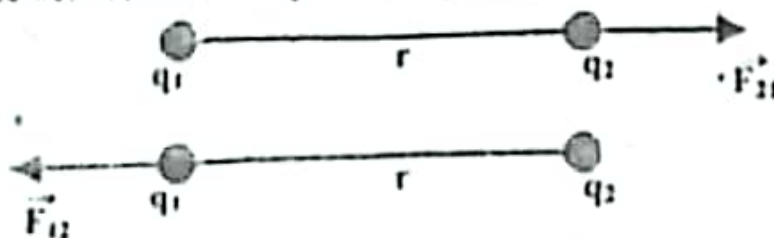
The Coulomb's force in free space is given as

$$F_e = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2}$$

The Coulomb's force is a mutual force. It means force $\vec{F}_{12}$ exerted by charge q_1 on charge q_2 is equal in magnitude but opposite in direction to force $\vec{F}_{21}$ exerted by charge q_2 on charge q_1 .

$$\vec{F}_{21} = -\vec{F}_{12}$$

The electric force between two charges in vector form is written as



EXAMPLES

Charge, energy of photons, frequency of vibrations of stretched string and angular momentum of electron in orbits are some examples of quantized physical quantities.

CHARGE CONSERVATION

A physical quantity is called conserved when total amount of physical quantity remains constant before change and after change.

EXAMPLES

Charge and energy are conserved in pair production and annihilation of matter.

Q:3 An electron charge $-e$ circulates around a helium nucleus of charge $+2e$ in helium atom. Which particle exerts the larger force on other?

Ans. The electric force between two charged bodies is action reaction force which is equal in magnitude but opposite in direction. The force exerted by electron on nucleus is equal to force exerted by nucleus on electron.

Q:4 Would it have any important difference if Benjamin Franklin had chosen, in effect, to call electrons positive and protons negative.

Ans. It makes no any important difference if Benjamin Franklin had chosen, in effect, to call electrons positive and protons negative.

Q:5 A solution of copper sulphate is a conductor. What particles serve as the charge carriers in this case.

Ans. The particles Cu^{++} and SO_4^{--} serve as charge carrier in solution of copper sulphate.

Q:6 Compare gravitational force and Coulomb force between two electrons. Which is greater?

Ans. The gravitational force between two electrons is

$$F_g = \frac{G m_e m_e}{r^2}$$

The Coulomb force between two electrons is

$$F_e = \frac{k q_1 \times q_2}{r^2} = k \frac{e \times e}{r^2}$$

The ratio is

$$\frac{F_e}{F_g} = \frac{k e^2}{r^2} \times \frac{r^2}{G m_e^2} = \frac{k e^2}{G m_e^2}$$

$$\frac{F_e}{F_g} = \frac{9 \times 10^9 (1.6 \times 10^{-19})^2}{6.67 \times 10^{-11} (9.1 \times 10^{-31})^2} = 4.17 \times 10^{42}$$

It shows that $F_e > F_g$

$$\vec{F}_{21} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2} \hat{r}_{21}$$

$$\vec{F}_{12} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2} \hat{r}_{12}$$

EFFECT OF MEDIUM

The presence of a medium called dielectric between charges reduces the electric force between them as compared to the force in free space by certain factor that is constant for the given dielectric.

This constant is called relative permittivity and denoted by ϵ_r . The values of relative permittivity of different dielectrics are given in table.

Material	Vacuum	Air(1atm)	Ammonia (liquid)	Bakelite	Benzene	Germanium	Glass
ϵ_r	1	1.0006	22-25	5-16	2.284	16	4.8-10
Material	Mica	Paraffined paper	Plexiglas	Rubber	Teflon	Transformer oil	Water (distilled)
ϵ_r	3-7.5	2	3.40	2.94	2.1	2.1	78.5

The Coulomb's force in medium of relativity permittivity ϵ_r is given as

$$F_{med} = \frac{F_e}{\epsilon_r} = \frac{1}{4\pi\epsilon_0 \epsilon_r} \frac{q_1 q_2}{r^2}$$

USES OF VECTOR FORM OF COLULOMB'S LAW

The vector form has following uses:

- 1 It tells whether force is attractive or repulsive.
- 2 The resultant force can be calculated from vector sum of forces when assembly of charges is under consideration.

ELECTRIC FORCE OF N-CHARGES

Consider assembly of n-charges. The total force exerted on charge "q" placed among n- point charges can be calculated by superposition principle as

$$\vec{F} = \vec{F}_1 + \vec{F}_2 + \vec{F}_3 + \dots + \vec{F}_n$$

$$\vec{F} = kq \sum_{i=1}^n \frac{q_i}{r_i^2} \hat{r}_i$$

Where $i = 1, 2, 3, 4, \dots, n$

SIGNIFICANCE OF COULOMB'S LAW

The Coulomb's law describes completely how electrons are attracted by nucleus in atom, how atoms form molecules and how atoms and molecules develop into solids or liquids.

The Coulomb's law not only holds for macroscopic charged particles, it is equally applicable to microscopic particles like electron, proton. However, law is not applicable when charged particles move close to velocity of light and inside the nucleus under certain conditions because there exist nuclear force rather than Coulomb's force.

Q.5. Explain how Coulomb's law is verified experimentally?

EXPERIMENTAL VERIFICATION

The charges of same polarity repel each other and charges of opposite polarity attract each other with a force called electrostatics force.

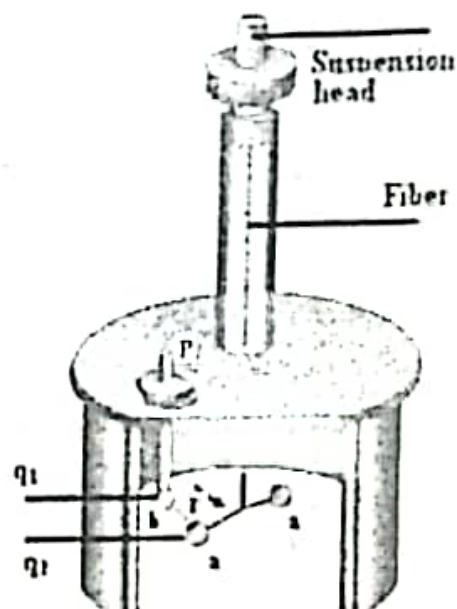
APPARATUS (TORSION BALANCE).

Coulomb measured electric force between charges using apparatus called torsion balance. The torsion balance consists of horizontal rod carrying two metallic spheres a and a' at its ends as shown in diagram.

The rod is suspended from the middle with a fiber. A small mirror M is attached to the fiber and beam of light reflected from mirror falls on the scale.

WORKING

An insulated rod P carrying a small charged sphere b equal in size to sphere a is brought near sphere a. Both spheres share charge $q_1 = q_2 = q$ when sphere b is touched with sphere a. The charge on sphere b can be further divided into $q/2$, $q/4$ by touching it with the spheres of equal sizes using same method.



By keeping the distance between sphere a and sphere b fixed, we find that twisting (electric force) in fiber is directly proportional to the magnitude of the charges on sphere a and sphere b.

The twisting (electric force) is found inversely proportional to square of the distance between the charged spheres by changing the distance between them.

SHORT QUESTIONS

Q:1 The quantum of charge is 1.6×10^{-19} C. Is there a corresponding quantum of mass?

Ans: There is no quantum of mass

Q:2 What does it mean to say that a physical quantity is (a) quantized or (b) conserved. Give some examples.

Ans: **CHARGE QUANTIZATION**

When a physical quantity exists in discrete packets, it is called quantized.

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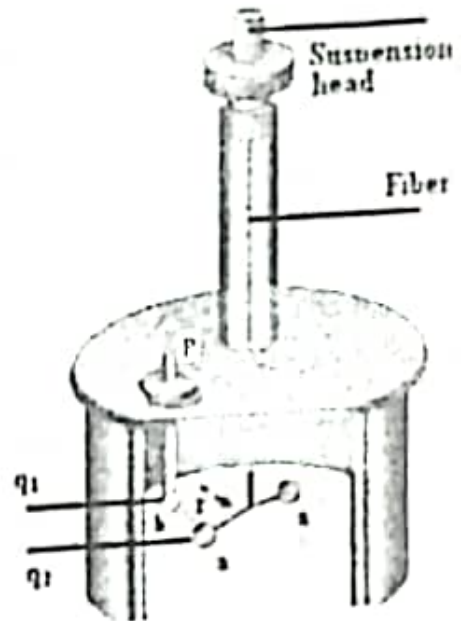
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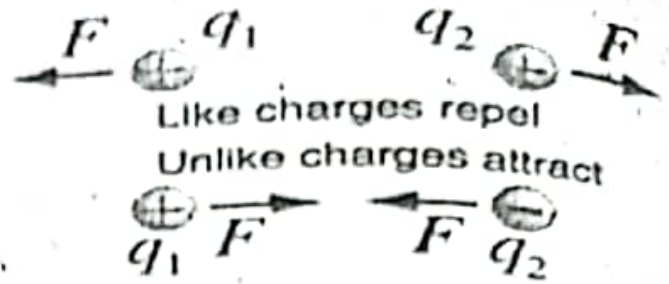
ELECTROSTATICS

The study of electric charges at rest under the action of electric forces is called electrostatics.

TYPES OF CHARGES

The charge has two types called positive charge and negative charge.

- ① The similar charges repel each other with electric force.
- ② The unlike charges attract each other with electric force.



POLARITY OF CHARGE

The charges of same polarity repel each other while charges of opposite polarity attract each other. The positive and negative labels to charged materials are arbitrary and assigned by the Franklin.

CHARGE ON ELECTRON

// The magnitude of charge on electron is 1.6×10^{-19} C. It is measured by Milliken.
The mass of an electron is 9.1×10^{-31} kg.

CHARGE ON PROTON

The magnitude of charge on proton is 1.6×10^{-19} C.

The mass of proton is 1.67×10^{-27} kg. //

CHARGE ON NEUTRON

The neutron has no charge. Its mass is approximately equal to mass of proton.

// POLES OF CHARGE

The charge forms poles.

- ① An isolated single charge is called an electric monopole.
- ② Equal positive and negative charges having constant separation between them is called electric dipole.
- ③ Two oppositely directed dipoles close to each other are called an electric quadrupole. //

You can continue this process to any number of poles, but dipoles and quadrupoles are important because they find significant application in physical phenomenon.

CONDUCTORS ..

The materials through which charges can flow easily are called conductors. Some properties of conductors are

- ① The copper and aluminum are good examples of conductors.
- ② The free electrons are mobile and positive ionic cores are immobile within conductor.

// Each atom of a typical conductor may contribute one conduction electron. It means a conductor on average has (10^{23}) conduction electrons/cm³.

- ① The conduction band and valence band overlaps in conductors so electrons can easily move.
- ② The resistivity of conductors is of the order of $10^{-8} \Omega\text{-m}$. The resistivity of Cu is $1.77 \times 10^{-8} \Omega\text{-m}$.

$$1.77 \times 10^{-8} \Omega\text{-m}$$

SEMICONDUCTORS

The partial conductors are called semiconductors.

- ① The well-known examples of semiconductors are silicon and germanium.

- ② A typical semiconductor has 10^{12} conduction electrons/cm³.

- ③ The resistivity of semiconductor germanium is $0.5 \Omega\text{-m}$.

Conductors / S/c / ρ in

10^{23}	10^{12}	or	10^{16}
10^{-8}	0.5		10^{-10}
			$\Omega\text{-m}$

INSULATORS

The materials through which charges cannot flow are called insulators.

- ① The electrons are tightly bound in such materials.
- ② The well-known insulators are glass and plastic.
- ③ The insulator has only one conduction electron/cm³.
- ④ The resistivity of insulators is of the order of 10^{11} — $10^{16} \Omega\text{-m}$. The resistivity of glass is $2 \times 10^{11} \Omega\text{-m}$.
- ⑤ The energy gap between conduction band and valence band in insulator is very high. The electrons cannot jump from valence band to conduction.

Q2 Discuss quantization of electric charge in detail?

QUANTIZATION OF CHARGE

The electric charge was thought to be a continuous fluid in Benjamin Franklin's day. Nowadays we know that fluids such as air and water are not continuous but are made of atoms and molecules. Experiment shows that electrical fluid is also not continuous but is made up of multiples of a certain elementary charge.

⇒ Any positive or negative charge q can be written as

$$q = ne \quad \checkmark$$

Where $n = \pm 1, \pm 2, \pm 3, \dots$ and $e = 1.6 \times 10^{-19} \text{ C}$

The elementary charge e is one of the important constants of nature. The electron and proton both have a charge of magnitude e . The quarks which are constituent particles of protons and neutrons have charges of $\pm \frac{e}{3}$ or $\pm \frac{2e}{3}$ but they cannot be detected individually. That is why we do not take their charges to be the elementary charge.

You often see phrases such as "the charge on a sphere," "the amount of charge transferred," and "the charge carried by the electron". These phrases suggest that charge is a substance. You should however keep in mind what is indeed. Particles are the substance and charge happens to be one of their properties just as mass is.

We say a physical quantity such as charge is quantized when it can have only discrete values rather than any value. For example, it is possible to find a particle that has no charge at all or a charge of $+10e$ or $-6e$ but not a particle with a charge of say $3.57e$.

The quantum of charge is small. For example, in an ordinary 100 W light bulb about 10^{19} elementary charges enter the bulb every second and just as many leave. However, the graininess of electricity does not show up in such large-scale phenomena (the bulb does not flicker with each electron) just as you cannot feel the individual molecules of water with your hand.

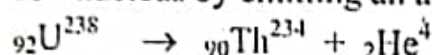
Q.3 Discuss conservation of electric charge in detail with examples?

CHARGE CONSERVATION

The positive charge appears on the rod when you rub a glass rod with silk. The measurement shows that negative charge of equal magnitude appears on the silk. This suggests that rubbing does not create charge but only transfers it from one body to another body. This hypothesis of conservation of charge was first put forward by Benjamin Franklin. Some examples of charge conservation are

RADIOACTIVE DECAY

The conservation of charge occurs in radioactive decay of nuclei in which a nucleus transforms into a different type of nucleus. For example, a uranium-238 nucleus transforms into a thorium-234 nucleus by emitting an alpha particle.

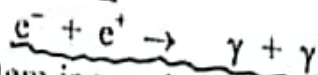


The parent nucleus ${}_{92}\text{U}^{238}$ contains 92 protons having charge of $+92e$. The daughter nucleus ${}_{90}\text{Th}^{234}$ contains 90 protons having charge of $+90e$. The emitted alpha particle ${}_2\text{He}^4$ contains 2 protons having charge of $+2e$. The total charge is $+92e$ before and after the decay. Thus charge is conserved.

The total number of protons and neutrons before decay are 238 and after decay are $234 + 4 = 238$. Hence total number of protons and neutrons are also conserved in decay process.

ANNIHILATION OF MATTER

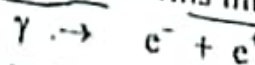
The charge conservation occurs in annihilation process when an electron having charge $-e$ and its antiparticle called positron having charge $+e$ annihilates in such a way that they transform into two gamma rays.



The net charge of the system is zero both before and after the event. Hence charge is conserved.

PAIR PRODUCTION

The charge is also conserved in pair production that is converse process of annihilation. In this process a gamma ray transforms into an electron and a positron



The net charge of the system is zero both before and after the event. Hence charge is conserved.